UNIT-9

Form Factor

&

Peak Factor

Of

Waveform

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(i) **Peak Value :** *The maximum value, positive or negative of an alternating quantity is known as peak value or maximum value or amplitude.* The peak value or maximum value of an alternating voltage or current is represented by E_m or I_m.

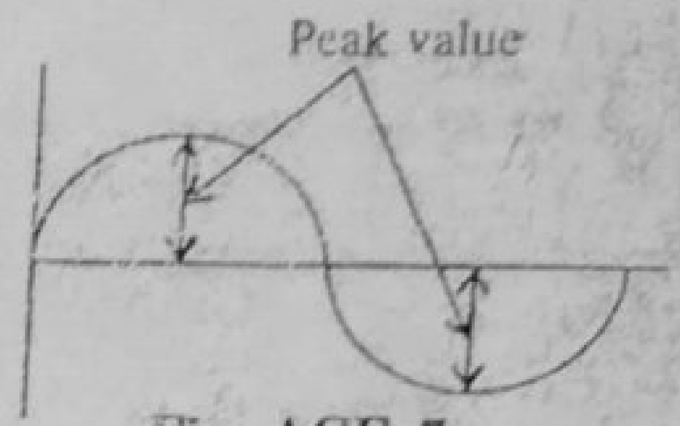

Fig. ACF-7

(ii) **Average Value :** *The average value of an alternating current I_{av} is expressed by that steady current which transfers across any circuit the same charge as is transferred by that alternating current.* In symmetrical waves (*i.e.* one whose two half cycles are exactly similar), the average value over a complete cycle is zero. However, the average value of a positive or negative half is not zero. Hence in the case of symmetrical waves, average value means the average value over a half cycle.

$$\therefore \text{ Average value of symmetrical wave} = \frac{\text{Area of one alternation(half cycle)}}{\text{Base length of one alternation}}$$

Average value of sinusoidal current : The equation of an alternating current varying sinusoidally is given by

$$i = I_m \sin\theta$$

Consider an elementary strip of thickness $d\theta$ in the first +ve half cycle in the wave as shown in fig.ACF-9. If i is the mid ordinate, then area of strip $= i d\theta$

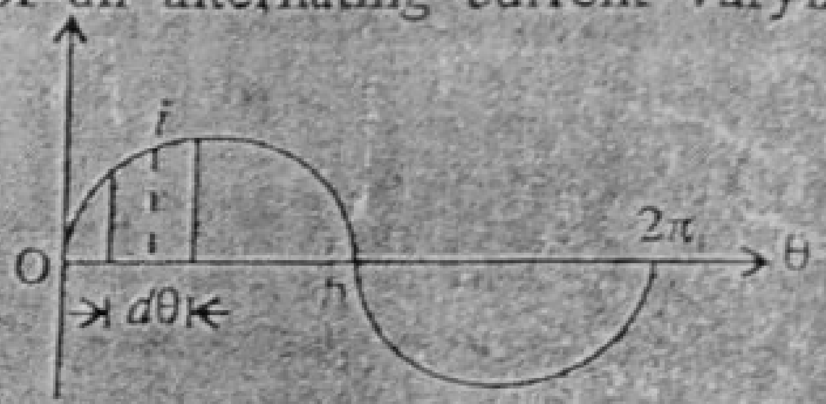

Fig. ACF-9

$$\text{Area of half cycle (+ ve)} = \int_0^\pi i \, d\theta = \int_0^\pi I_m \sin\theta \, d\theta$$

$$= I_m[-\cos\theta]_0^\pi = 2I_m$$

$$\therefore \text{ Average value of current, } I_{av} = \frac{\text{Area of alternation}}{\text{Base}}$$

or
$$I_{av} = \frac{2I_m}{\pi} = 0.637\, I_m$$

Average value of current = 0·637 × Maximum value

$$I_{av} = 0.637\, I_m$$

(iii) **Effective value or R.M.S Value :–** *R. M.S. value of an alternating current is given by that steady current (d.c.) which when flowing through a given resistance for a given time produces the same heat as produced by alternating current when flowing through the same resistance for the same time.* This is also known as effective or virtual value of *A.C.*

The *r.m.s.* value of symmetrical wave can be expressed as :

$$R.M.S.\text{ value} = \sqrt{\frac{\text{Area of half cycle of squared wave}}{\text{half cycle base}}}$$

or

Similarly the *r.m.s.* value of the alternating voltage is given by the expression

$$V = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots\dots + v_n^2}{n}}$$

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Analytical Method : The standard equation of the alternating current varying sinusoidally is given by

$$i = I_m \sin\theta$$

Consider an elementry strip of thickness "$d\theta$" in first half cycle of squared wave.

Let i^2 be the mid-ordinate of the strip,

$\therefore$ Area of the strip $= i^2 \cdot d\theta$.

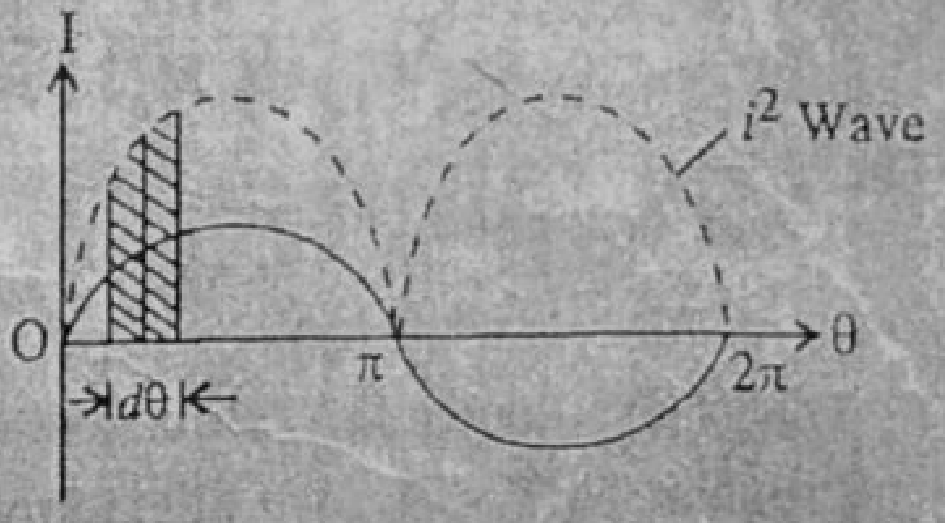

Fig. ACF-11

$$\text{Area of half cycle} = \int_0^{\pi} i^2 . d\theta$$

$$= \int_0^{\pi} I_m^2 . \sin^2\theta \, d\theta$$

$$= I_m^2 \int_0^{\pi} \sin^2\theta \, d\theta = I_m^2 \int_0^{\pi} \frac{(1-\cos 2\theta)}{2} d\theta$$

$$\frac{I_m^2}{2}\int_0^{\pi}(1-\cos 2\theta)d\theta = \frac{I_m^2}{2}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi}$$

$$= \frac{I_m^2}{2}\left[(\pi - 0) - \frac{\sin 2\pi - \sin 0}{2}\right]$$

$$= \frac{I_m^2}{2}\left[(\pi - 0) - (0 - 0)\right] = \frac{I_m^2}{2}\cdot\pi$$

Base 0 to $\pi = \pi - 0 = \pi$

$$\therefore \quad R.M.S.\text{ value} = \sqrt{\frac{\text{Area of first half of squared wave}}{\text{Base}}}$$

$$= \sqrt{\frac{\pi . I_m^2}{2\pi}} = \frac{I_m}{\sqrt{2}} = 0.707\, I_m$$

$$\therefore \quad I_e = 0.707\, I_m$$

Form factor. *It is defined as the ratio of RMS value to the average value* of an alternating quantity. *It is denoted by* K_f.

i.e. $$K_f = \frac{\text{R.M.S. value}}{\text{Average value}}$$

For sinusoidal alternating currents, $K_f = \frac{0.707 I_m}{0.637 I_m} = 1.1$ = form factor for sinusoidal a.c.

The value of form factor K_f depends upon the wave shape of the alternating quantity. Its least value is 1 and may be as high as 5 for other wave shapes. However for sinusoidal voltage or current,

$$\text{Form factor } K_f = \frac{0.707\times\text{Max. value}}{0.637\times\text{Max. value}} = 1.11 = \frac{I_{rms}}{I_{avg}} = \frac{E_{rms}}{E_{avg}} = \frac{I_{rms}}{I_{avg}}$$

Peak Factor. *It is defined as the ratio of maximum value to R.M.S. value. It is denoted by* K_p

i.e. $$K_p = \frac{\text{Max. value}}{\text{R.M.S. value}}$$

$$\text{Peak factor} = K_p = \frac{I_m}{I_{r.m.s}} \quad \text{or} \quad K_p = \frac{E_m}{E_{r.m.s.}}, \; K_p = \frac{I_m}{\frac{I_m}{\sqrt{2}}} = \sqrt{2} = 1.4142$$ (for sinusoidal alternating currents)

This is also known as *crest* or *amplitude factor.*

For sinusoidal alternating voltage, $K_p = \frac{E_m}{E_m/\sqrt{2}} = 1.414$ = Peak factor

Numericals based on Peak factor and Form factor calculations:

100π 3 300

Problem ACF–7. A current has the following steady values in amperes for equal intervals of time changing instantaneously from one value to another *i.e.* to the next.

0, 20, 40, 60, 40, 20, 0, –20, –40, –60, –40, –20, 0

Calculate *(i)* Average value *(ii)* R.M.S value *(iii)* Form factor *(iv)* Peakfactor.

Sol. The figure shows the wave form.

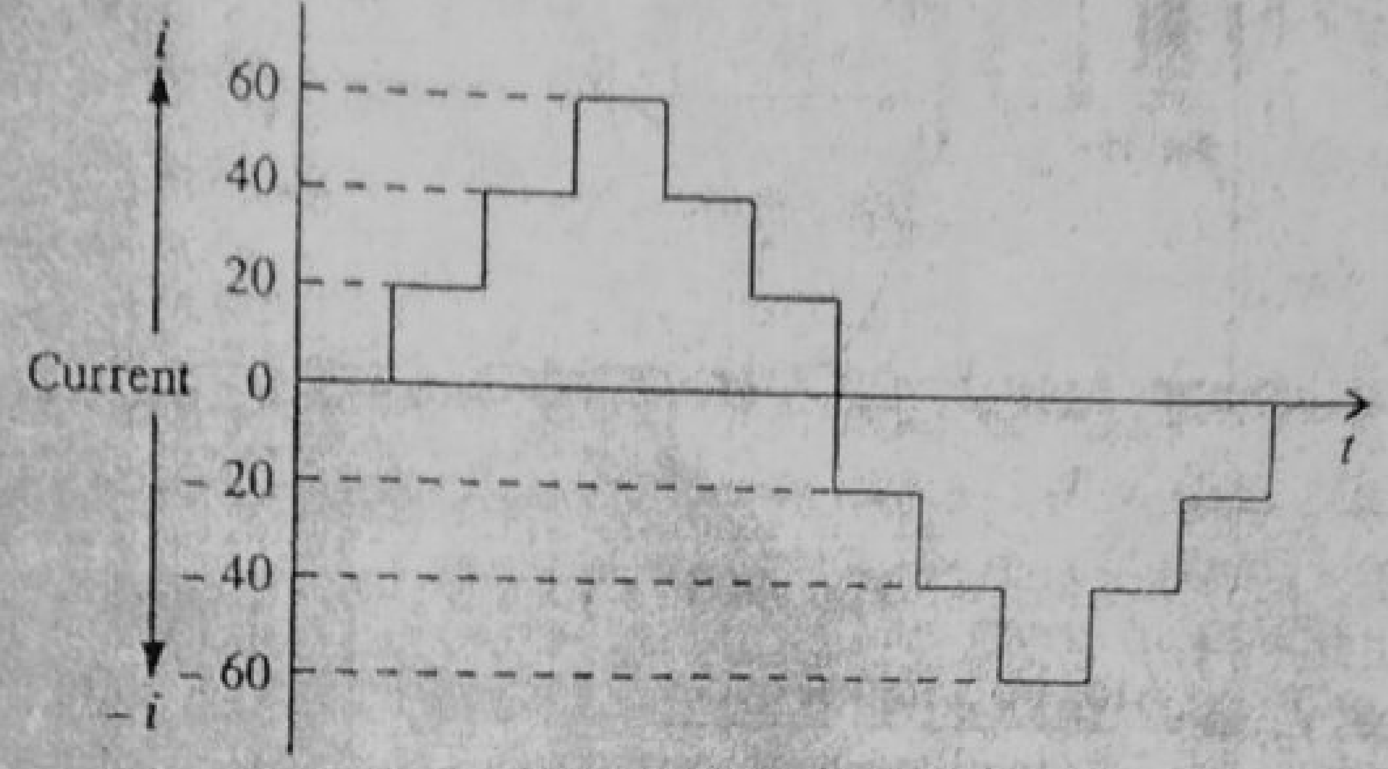

Fig. ACF – 12

(i) **Average value of current,** $I_{av} = \dfrac{i_1 + i_2 + i_3 + i_4 + i_5 + i_6}{6}$

$$= \frac{0 + 20 + 40 + 60 + 40 + 20}{6} = 30 \text{ A (Ans.)}$$

(iii) **R.M.S. value of current**

$$I^2_{rms} = \frac{i_1^2 + i_2^2 + i_3^2 + i^2_4 + i^2_5 + i^2_6}{6}$$

$$= \frac{0^2 + 20^2 + 40^2 + 60^2 + 40^2 + 20^2}{6}$$

$$= \frac{0 + 400 + 1600 + 3600 + 1600 + 400}{6}$$

$$= \frac{7600}{6} = 1266{\cdot}67 \text{ A}$$

$$I_{r.m.s.} = \sqrt{1266 \cdot 67} = 35{\cdot}59 \text{ A (Ans.)}$$

(iii) Form factor, $K_f = \dfrac{35 \cdot 59}{30} = 1{\cdot}186$ (Ans.)

(iv) Peak factor, $K_p = \dfrac{60}{35 \cdot 59} = 1{\cdot}685$ (Ans.)

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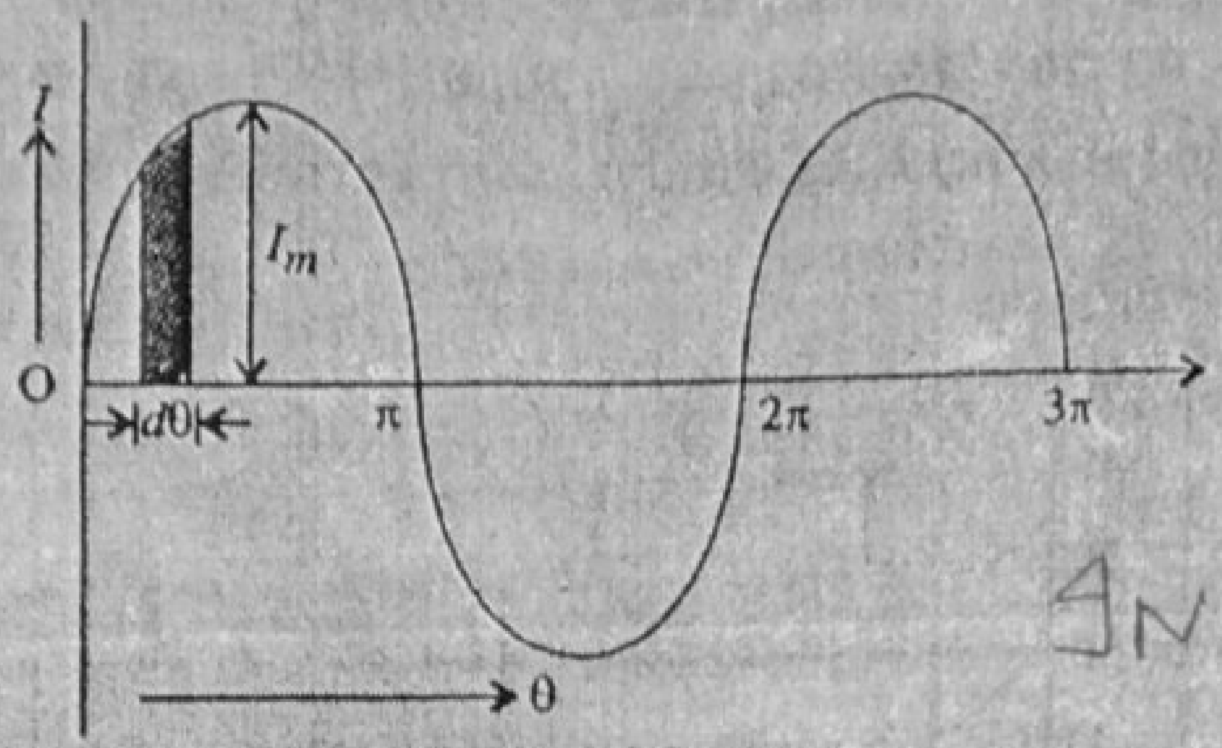
Problem ACF – 8. For a half wave rectified alternating current, find *(i)* Average value *(ii)* R.M.S. value *(iii)* Form factor *(iv)* Peak factor.

Sol. Half wave (H.W.) rectified alternating current is one, whose one half cycle has been supressed *i.e.* current flows only during positive half cycle.

The wave is symmetrical, therefore, one complete cycle is considered.



INPUT (Pure Sine Wave)

Fig. ACF – 13

(i) Average value, $I_{av} = \dfrac{\text{Area of complete cycle}}{\text{Base}}$

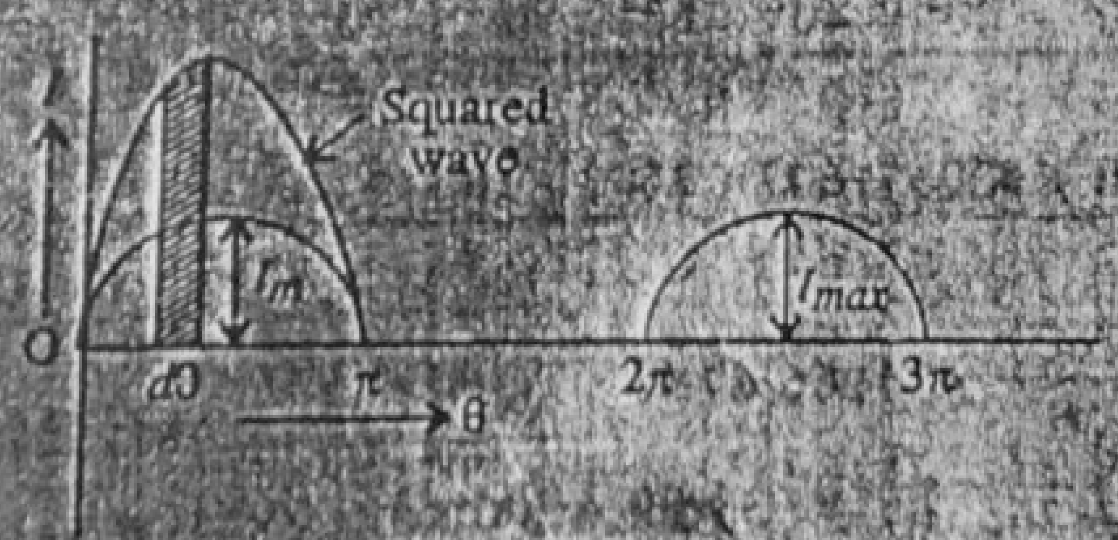
Now Area of complete cycle = Area of half cycle ($\because$ –ve half is suppressed)

$$= \int_0^\pi i\,d\theta = \int_0^\pi I_m \sin\theta\, d\theta = I_m[-\cos\theta]_0^\pi = 2\,I_m$$

Base = 2 π (*i.e.* 0 to 2 π or 2 π – 0 = 2 π) Base = 2π

$$\therefore \quad I_{av} = \frac{2I_m}{2\pi} = \frac{I_m}{\pi} = \frac{I_m}{3.142} = 0.3183\ I_m \text{ (Ans.)}$$

(ii) For finding *r.m.s.* value, the squared wave of a half wave rectified wave is shown in fig. ACF –14.



HWR OUTPUT

Fig. ACF–14

Area of a squared wave over a complete cycle

= Area of a squared wave in half cycle

(Since negative half cycle is suppressed)

RMS Value of Half-Wave rectified output

$i = I_m \sin\theta\, d\theta \qquad 0 \le \omega t \le 180^\circ$

$i = 0 \qquad 180^\circ \le \omega t \le 360^\circ$

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$$= \int_0^{2\pi} i^2 d\theta = \int_0^{2\pi} I_m^2 \sin^2\theta \, d\theta = \int_0^{\pi} I_m^2 \sin^2\theta \, d\theta + \int_{\pi}^{2\pi} 0 \cdot d\theta \quad (\because i = I_m \sin\theta) \quad \left[\text{for } 0 \le \omega t \le \pi\right]$$

$$= I_m^2 \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2}\right) . d\theta = \frac{\pi \cdot I_m^2}{2}, \quad \text{Base} = 2\pi$$

$$\therefore \quad I_{r.m.s.} = \sqrt{\frac{\pi I_m^2}{2 \times 2\pi}} = \frac{I_m}{2} \text{ (Ans.)}$$

(iii) Form factor, $K_f = \frac{I_{rms}}{I_{av}} = \frac{I_m}{2} \times \frac{\pi}{I_m} = 1{\cdot}57$ (Ans.)

(iv) Peak factor, $K_p = \frac{I_m}{I_{r.m.s.}} = \frac{I_m}{I_m} \times 2 = 2$ (Ans.)

Questions for Practice:

Ques 1) Calculate form factor and peak factor of :

$$v = V_m \cos(\omega t)$$

Ques 2) Compute form factor and peak factor of :

$$v = V_m \sin(2\omega t)$$

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